



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (E&E) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EE303 - DIGITAL SYSTEM DESIGN

20EE305 - LOGIC DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Determine prime implicants, essential prime implicants and minimal sum for the given function $f(w,x,y,z) = \sum m(4,5,6,7,8,11,12,13,15) + \sum d(3,9)$ using Karnaugh Map technique.	6	L*3	1	1
	b) For a given Boolean function determine minimal sum using MEV technique taking d as the map entered variable. $f(a,b,c,d) = \sum m(2,3,4,5,13,15) + \sum d(8,9,10,11)$	6	L3	1	1
	c) With the help of neat diagram derive logic expressions for the carry outputs of a 4-stage carry look-ahead adder.	8	L3	2	1
2.	a) Using K-map technique find minimal product for the given function. $f(w,x,y,z) = \prod M(0,2,3,5,7)$	4	L2	1	1
	b) Design 2:4 Decoder with active low output with the help of truth table and necessary sketches.	8	L3	2	1
	c) Implement the following functions using 3:8 Decoder considering minimum number of inputs. $f1 = \sum(0,1,5,6,7)$ $f2 = \prod m(0,3,5,6,7)$	8	L3	2	1
3.	a) Explain 4x2 priority encoder with the help of neat diagrams and truth table.	8	L3	2	1
	b) Implement $f(a,b,c,d) = \sum m(1,3,4,11,12,13,14,15)$ using 8 to 1 MUX using a, b and c as the address lines.	8	L3	2	1
	c) Simply $f(x,y,z) = \sum(0,2,3,7)$ and get the expressions of essential prime implicants and minimal sum.	4	L2	1	1
Unit – II					
4.	a) Explain Master-Slave JK flipflop with the help of neat sketches and timing diagram.	10	L2	3	1
	b) Differentiate between latches and flipflops. With the help of neat sketch and timing diagram explain Gated SR Latch.	10	L2	3	1
5.	a) Explain positive edge triggered D-flipflop with the help of necessary diagrams and truth table.	10	L3	3	1
	b) With a neat logic diagram explain the following i) Mod-4 ring counter ii) 4-bit synchronous down counter using AND gates	10	L3	4	1
6.	a) Draw the logic diagram and truth table using NAND gates for i) SR flipflop ii) D flipflop	10	L1	3	1

P.T.O.

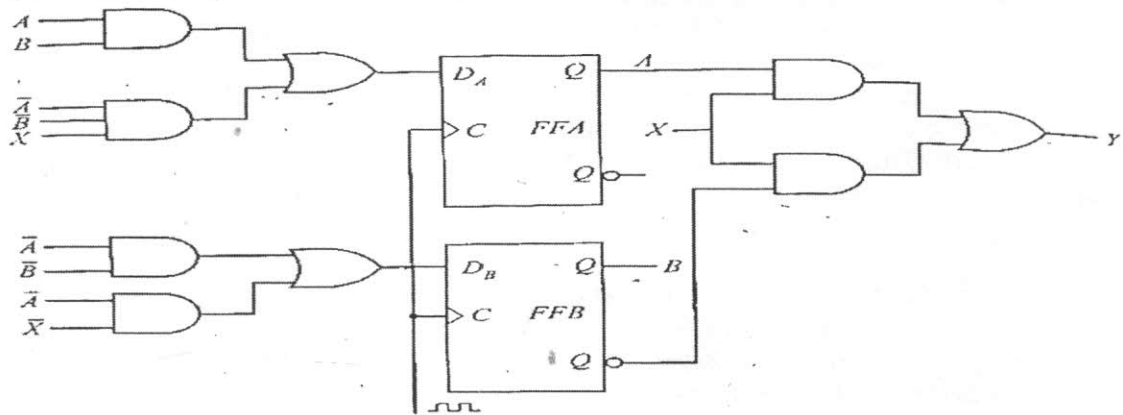
- b) Design a Universal shift register incorporating four multiplexers and four positive edge triggered D flipflops having states specified in the table below.

Select lines		Register operations
S1	S0	
0	0	Hold
0	1	Shift right
1	0	Shift left
1	1	Parallel load

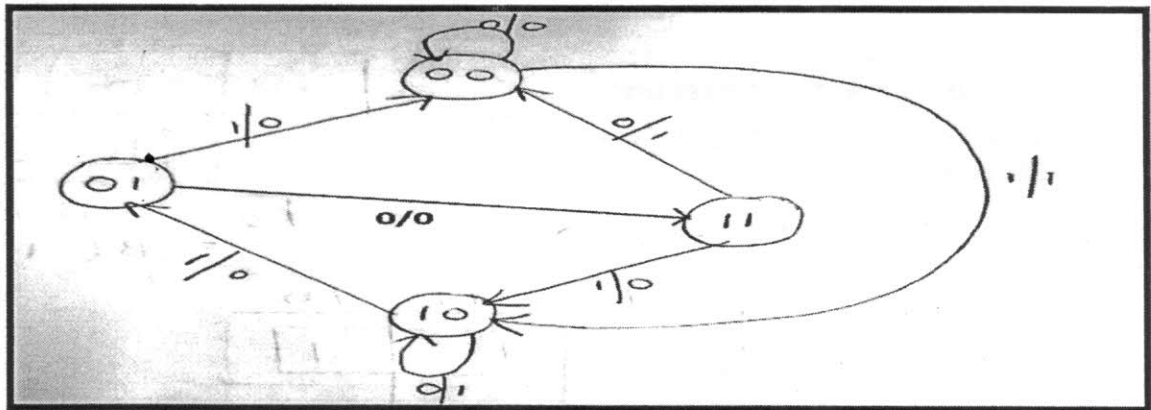
10 L3 4 1

Unit – III

7. a) Construct the excitation table, transition table, state table and state diagram for the sequential circuit below.

10 L3 5 1
10 L2 5 1

8. a) Construct a sequential circuit from the state diagram using clocked T flipflops.



10 L3 5 1

- b) Configure a n- channel enhancement type MOSFETS to perform the Boolean NAND and NOR function.

10 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
